Class 11 CBSE Computer Science

Chapter 8 – Strings

Questions First, Answers Later

Based on NCERT + CBSE Syllabus

PART A – QUESTIONS

Very Short / Method Explanation Questions

1. Explain the `capitalize()` method in Python.
2. Write the syntax of the `capitalize()` method.
3. What does the `count()` method do? Write its general form.
4. What does the `endswith()` method return?
5. Write the syntax of the `find()` method.
6. Write the syntax of the `isalnum()` method.
7. Write the syntax of the `isalpha()` method.
8. Describe the `isdigit()` method.
9. Why do we use the `islower()` method in Python?
10. Describe the `isspace()` method.
11. What is the use of the `strip()` method? Write its syntax.
12. What is a string in Python?
13. What is indexing in the context of Python strings? Why is it called two-way indexing?
14. What is a string slice? How is it useful?
15. How can you "update" an existing string in Python?
16. Describe Triple Quotes in Python.
17. Define a raw string with an example.

Output Prediction Questions

18. Write the output of the following code:

```
str1 = "this is string example....wow!!!"
str2 = "exam"
print(str1.find(str2))
print(str1.find(str2, 10))
print(str1.find(str2, 40))
```

19. Write the output of the following code:

```
str = "this2009"
print(str.isalnum())
str = "this is string example....wow!!!"
print(str.isalnum())
```

20. Write the output of the following code:

```
str = "this"
print(str.isalpha())
str = "this is string example....wow!!!"
print(str.isalpha())
```

21. Write the output of the following code:

```
str = " "
print(str.isspace())
str = "This is string example....wow!!!"
print(str.isspace())
```

22. Write the output of the following code:

```
str = "this is string example....wow!!!"
print(str.upper())
```

23. Rectify the error (if any) in the given statements:

```
>>> str = "Hello World"
>>> str[5] = 'p'
```

24. Give the output of the following statements:

```
>>> str = "Honesty is the best policy"
>>> str.replace('o', '*')
```

25. Write the output of the following code:

```
str = "this is string example....wow!!!"
print(str.title())
```

26. Write the output of the following code:

```
str = "this is string example....wow!!!"
print(str.swapcase())
str = "THIS IS STRING EXAMPLE....WOW!!!"
print(str.swapcase())
```

— End of Questions —

Answers start on the next page

PART B – ANSWERS

Answers to Method Explanation Questions

1. The `capitalize()` method returns a copy of the string with only its first character converted to uppercase and the rest of the characters in lowercase.

2. Syntax: `str.capitalize()`

3. The `count()` method returns the number of occurrences of a substring in the given string (optionally within a specified range).
   General form: `str.count(sub[, start[, end]])`

4. The `endswith()` method returns `True` if the string ends with the specified suffix, otherwise it returns `False`.

5. Syntax: `str.find(sub[, start[, end]])`

6. Syntax: `str.isalnum()`

7. Syntax: `str.isalpha()`

8. The `isdigit()` method returns `True` if all characters in the string are digits, otherwise `False`.

9. The `islower()` method checks whether all the case-based characters (letters) of the string are in lowercase. It returns `True` if they are, otherwise `False`.

10. The `isspace()` method returns `True` if the string consists only of whitespace characters, otherwise `False`.

11. The `strip()` method returns a copy of the string with leading and trailing characters removed (default is whitespace).
    Syntax: `str.strip([chars])`

12. A string in Python is a sequence of Unicode characters enclosed in single quotes (`' '`), double quotes (`" "`) or triple quotes (`''' '''` or `""" """`).

13. Indexing means assigning a unique position number to each character of a string.
    It is called **two-way indexing** because:

    - Forward direction: indices start from 0, 1, 2, . . . , len-1
    - Backward direction: indices start from -1, -2, -3, . . . , -len

14. A string slice is a sub-part of a string obtained using the slicing operator `s[start:end]`. It returns characters from index `start` up to (but not including) `end`.
    Example: `"Computer"[1:4]` gives `"omp"`

15. Strings are immutable. To "update" a string, we create a new string by re-assigning the variable, often using slicing and concatenation.
    Example:

```
var1 = "Hello World!"
print("Updated String:-", var1[:6] + "Python")
```

16. Triple quotes (''' ''' or """ """) allow a string to span multiple lines and include newlines, tabs and special characters without escape sequences.

17. A raw string treats the backslash (\) as a normal character.
Example:

```
print(r'C:\nowhere')    # Output: C:\nowhere
```

Answers to Output Prediction Questions

18.
```
15
15
-1
```

19.
```
True
False
```

20.
```
True
False
```

21.
```
True
False
```

22.
```
THIS IS STRING EXAMPLE....WOW!!!
```

23. Strings are immutable, so individual characters cannot be changed.
Correct way:

```
s = list("Hello World")
s[5] = 'p'
print(''.join(s))    # Hello porld
```

24.
```
H*nesty is the best p*licy
```

25.
```
This Is String Example....Wow!!!
```

26.
```
THIS IS STRING EXAMPLE....WOW!!!
this is string example....wow!!!
```